

UNIT V NUCLEAR REACTIONS

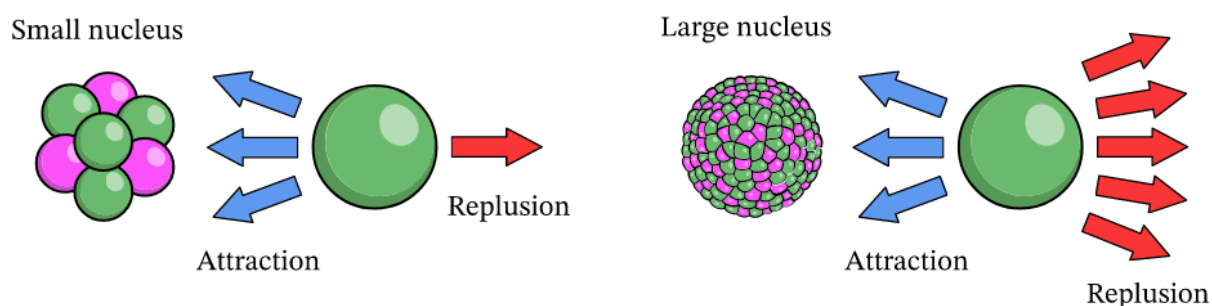
Lesson Plan 5.2

Syllabus: Nuclear stability- Patterns of nuclear stability - Variation in properties between isotopes

Reference: Chemistry for Engineering students by Lawrence S Brown and Thomas A Holme (Fourth Edition) Chapter 13

Introduction

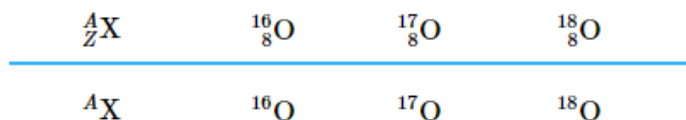
Although most of the known elements have at least one isotope whose atomic nucleus is stable indefinitely, all elements have isotopes that are unstable and disintegrate, or decay, at measurable rates by emitting radiation. Some elements have no stable isotopes and eventually decay to other elements. In contrast to the chemical reactions that were the main focus of earlier chapters and are due to changes in the arrangements of the valence electrons of atoms, the process of nuclear decay results in changes inside an atomic nucleus. We begin our discussion of nuclear reactions by reviewing the conventions used to describe the components of the nucleus.



The Atomic Nucleus

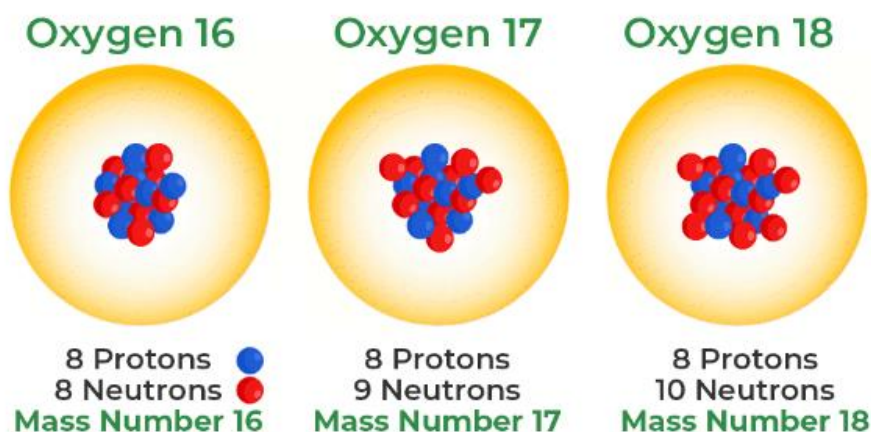
Each element can be represented by the notation ${}^A_Z\text{X}$, where A, the mass number, is the sum of the number of protons and the number of neutrons, and Z, the atomic number, is the number of protons. The protons and neutrons that make up the nucleus of an atom are called nucleons, and an atom with a particular number of protons and neutrons is called a nuclide. Nuclides with the same number of protons but different numbers of neutrons are called isotopes. Isotopes can also be represented by an alternative notation that uses the name of the element followed by the mass number, such as carbon-12. The stable isotopes of oxygen, for example, can be represented in any of the following ways:

stable isotopes of oxygen represented in different ways



element-A: oxygen-16 oxygen-17 oxygen-18

Because the number of neutrons is equal to $A - Z$, we see that the first isotope of oxygen has 8 neutrons, the second isotope 9 neutrons, and the third isotope 10 neutrons. Isotopes of all naturally occurring elements on Earth are present in nearly fixed proportions, with each proportion constituting an isotope's natural abundance.

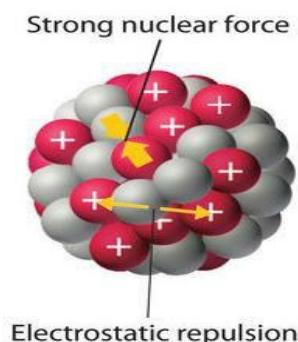


For example, in a typical terrestrial sample of oxygen, 99.76% of the O atoms is oxygen-16, 0.20% is oxygen-18, and 0.04% is oxygen-17. Any nucleus that is unstable and decays spontaneously is said to be radioactive, emitting subatomic particles and electromagnetic radiation. The emissions are collectively called radioactivity and can be measured. Isotopes that emit radiation are called radioisotopes.

Nuclear Stability

1. The Nucleus and its Composition:

- The nucleus of an atom is extremely small compared to the total volume of the atom.
- It contains protons and neutrons (nucleons), which define the isotope of an element.
- Despite its tiny size, it holds nearly all the mass of the atom.



2. Electrostatic Repulsion:

- Since protons are positively charged, they repel each other due to electrostatic forces.
- Without a counteracting force, the nucleus would disintegrate due to this repulsion.

3. The Strong Nuclear Force:

- The strong nuclear force is an attractive force between nucleons (protons and neutrons).
- It overcomes electrostatic repulsion and binds the nucleus together.
- This force is short-range, meaning it acts only over distances within the nucleus.

4. Neutrons as Nuclear Stabilizers:

- Neutrons act as "glue" between protons, reducing electrostatic repulsion.
- Hydrogen-1 (^1H) is the only stable nucleus without neutrons.
- As the number of protons increases, more neutrons are required for stability.
- Example: Helium-4 (^4He) has 2 protons and 2 neutrons, making it very stable.

Neutron-to-Proton Ratio and Nuclear Stability

1. The Band of Stability:

- A plot of neutron number (N) vs. proton number (Z) forms a band of stability where stable isotopes exist.
- Lighter elements ($Z \leq 20$): Stable isotopes have nearly equal numbers of protons and neutrons ($N/Z \approx 1$).
- Heavier elements require more neutrons to maintain stability ($N/Z \approx 1.5$ for the heaviest stable elements).

2. Instability Due to Imbalance:

- If N/Z is too low or too high, the nucleus becomes unstable and undergoes radioactive decay.
- Elements beyond $Z = 83$ (Bismuth) are all unstable and decay over time.

3. Decay Processes Based on N/Z Ratio:

- Neutron-rich nuclei (high N/Z): Undergo beta decay (β^- emission), converting a neutron into a proton.
- Proton-rich nuclei (low N/Z): Undergo positron emission (β^+ decay) or electron capture, converting a proton into a neutron.
- Very heavy nuclei: Undergo alpha decay (α -emission) to reduce both protons and neutrons.

4. Examples of Stable and Unstable Isotopes:

- Stable: ^{12}C (carbon-12), ^{16}O (oxygen-16), ^{56}Fe (iron-56)
- Unstable: ^{14}C (carbon-14, beta decay), ^{238}U (uranium-238, alpha decay), ^{206}Bi (bismuth-206, radioactive decay)

Patterns of Stability: Magic Numbers and Nuclear Shells

1. Magic Numbers:

- Certain proton or neutron numbers create highly stable nuclei.
- These are called magic numbers: 2, 8, 20, 28, 50, 82, 126.
- Example: Lead-208 (^{208}Pb) is doubly magic (82 protons, 126 neutrons) and highly stable.

2. Doubly Magic Nuclei:

- If both proton and neutron numbers match magic numbers, the nucleus is exceptionally stable.
- Examples:
 - Helium-4 (^4He): 2 protons, 2 neutrons.
 - Oxygen-16 (^{16}O): 8 protons, 8 neutrons.
 - Lead-208 (^{208}Pb): 82 protons, 126 neutrons (heaviest stable isotope).

3. Analogy to Electron Shells:

- Just like electrons arrange in shells in atoms, nucleons (protons & neutrons) fill nuclear shells. This explains why nuclei with magic numbers are more stable, similar to noble gases in the periodic table.

The "Peninsula of Stability" and the "Sea of Instability"

Regions of Nuclear Stability:

- Stable nuclei form a "peninsula of stability" surrounded by unstable isotopes in a "sea of instability."
- Beyond $Z > 83$ (Bismuth), all elements are unstable and eventually decay.

Superheavy Elements and the Island of Stability:

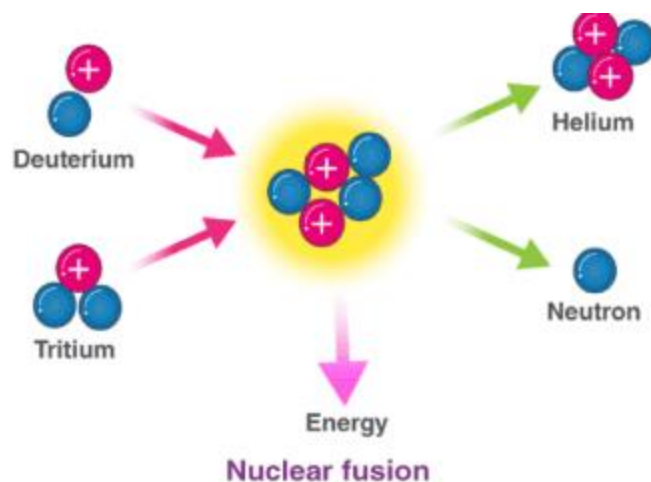
- Theoretical models predict a small region, called the "island of stability," where superheavy elements ($Z \approx 114-126$) might have long half-lives.
- Some experiments suggest Element 114 (Flerovium, Fl) has a longer-than-expected lifetime, supporting this idea.

Competing interactions within the atomic nucleus. Electrostatic repulsions between positively charged protons would normally cause the nuclei of atoms (except H) to fly apart. In stable atomic nuclei, these repulsions are overcome by the strong nuclear force, a short-range but powerful attractive interaction between nucleons. If the attractive interactions due to the strong nuclear force are weaker than the electrostatic repulsions between

protons, the nucleus is unstable, and it will eventually decay.

The relationship between the number of protons and the number of neutrons in stable nuclei, arbitrarily defined as having a half-life longer than 10 times the age of Earth, is shown graphically in Figure. The stable isotopes form a “peninsula of stability” in a “sea of instability.” Only two stable isotopes, ^1H and ^3He , have a neutron-to-proton ratio less than 1. Several stable isotopes of light atoms have a neutron-to-proton ratio equal to 1.

All other stable nuclei have a higher neutron-to-proton ratio which increases steadily to about 1.5 for the heaviest nuclei. Regardless of the number of neutrons, however, all elements with $Z > 83$ are unstable and radioactive. The Relationship between Nuclear Stability and the Neutron-to-Proton Ratio. In this plot of the number of neutrons versus the number of protons, each black point corresponds to a stable nucleus. In this classification, a stable nucleus is arbitrarily defined as one with a half-life longer than 46 billion years (10 times the age of Earth). As the number of protons (the atomic number) increases, the number of neutrons required for a stable nucleus increases even more rapidly. Isotopes shown in red, yellow, green, and blue are progressively less stable and more radioactive; the farther an isotope is from the diagonal band of stable isotopes, the shorter its half-life. The purple dots indicate superheavy nuclei that are predicted to be relatively stable, meaning that they are expected to be radioactive but to have relatively long half-lives. In most cases, these elements have not yet been observed or synthesized.



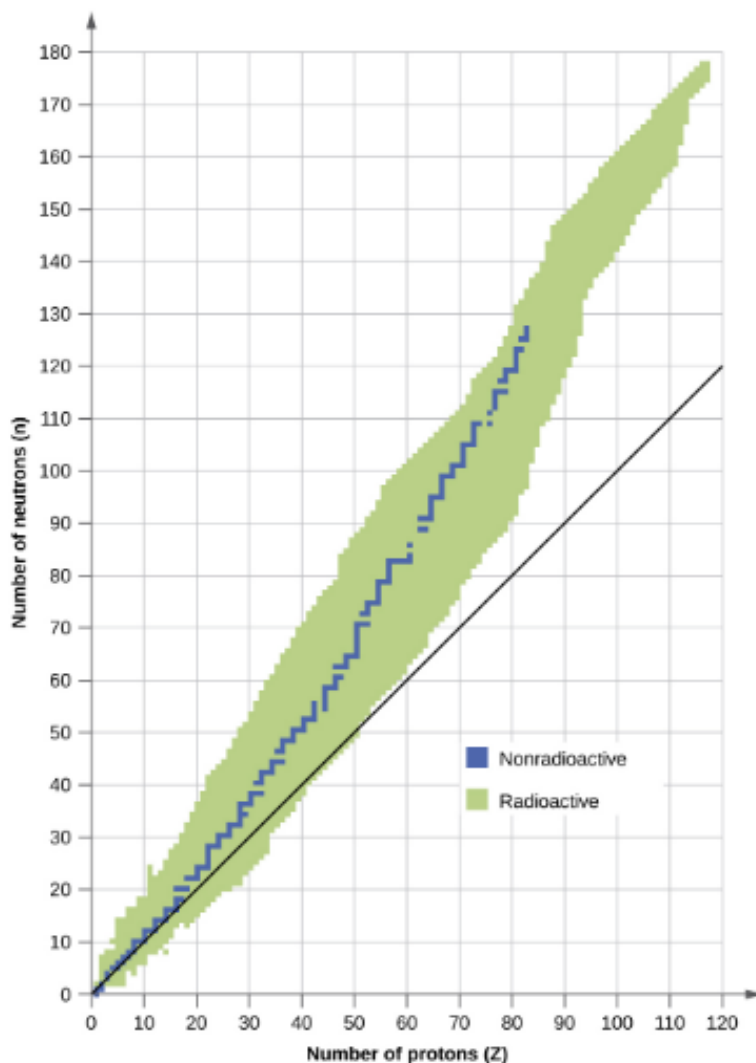


Figure. Graph of number of neutrons against the number of protons. The graph is divided into sections of "sea of instability", "peninsula of stability", "sea of instability" and "island of stability"

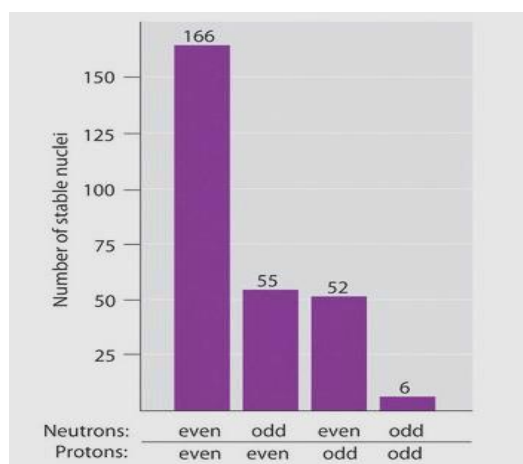
As shown in Figure, more than half of the stable nuclei (166 out of 279) have *even* numbers of both neutrons and protons; only 6 of the 279 stable nuclei do not have odd numbers of both. Moreover, certain numbers of neutrons or protons result in especially stable nuclei; these are the so-called *magic numbers* 2, 8, 20, 50, 82, and 126

For example, tin ($Z = 50$) has 10 stable isotopes, but the elements on either side of tin in the periodic table, indium ($Z = 49$) and antimony ($Z = 51$), have only 2 stable isotopes each. Nuclei with magic numbers of *both* protons *and* neutrons are said to be “doubly magic” and are even more stable. Example of elements with doubly magic nuclei are He, with 2 protons and 2 neutrons, and 208 with 82 protons and 126 neutrons, which is the heaviest known stable isotope of any element.

Stable Nuclear Isotopes

Number of Stable Isotopes	Proton Number	Neutron Number
157	even	even
53	even	odd
50	odd	even
5	odd	odd

The pattern of stability suggested by the magic numbers of nucleons is reminiscent of the stability associated with the closed-shell electron configurations of the noble gases in group 18 and has led to the hypothesis that the nucleus contains shells of nucleons that are in some ways analogous to the shells occupied by electrons in an atom. As shown in Figure, the “peninsula” of stable isotopes is surrounded by a “reef” of radioactive isotopes, which are stable enough to exist for varying lengths of time before they eventually decay to produce other nuclei.



Most stable nuclei contain **even** numbers of both neutrons and protons

Origin of the Magic Numbers

Multiple models have been formulated to explain the origin of the magic numbers and two

popular ones are the Nuclear Shell Model and the Liquid Drop Model.

Super heavy Elements

1. Experimental Evidence for Super heavy Elements

- Discovery of elements with $Z = 114$ (Flerovium) and $Z = 116$ (Livermorium) provided evidence supporting the existence of an island of stability.
- In 1999, Russian and American scientists synthesized a few atoms of element **114**, and one isotope had a half-life of 2.7 seconds, much longer than expected for such a heavy nucleus.
- This was considered an "eternity" in nuclear chemistry because most superheavy elements decay in microseconds or milliseconds.
- Further experiments at Dubna (Russia), Lawrence Livermore National Laboratory (USA), and GSI Helmholtz Centre for Heavy Ion Research (Germany) confirmed more superheavy elements up to Oganesson ($Z = 118$).

4. Long-Lived Super heavy Isotopes

- Some evidence suggests that an isotope with $A = 292$ may have been detected in thorium (^{232}Th) samples.
- If confirmed, this would indicate the possible natural existence of superheavy nuclei.
- The isotope's estimated half-life is greater than 10^8 years, suggesting exceptional stability compared to other synthetic superheavy elements.

5. Implications for Nuclear Physics

- The discovery of long-lived superheavy elements could reshape our understanding of nuclear structure and stability.
- It may allow new chemical experiments, exploring how these elements behave in reactions.
- Some theorists suggest that elements beyond $Z = 126$ could exist and remain stable for extended periods.

6. Challenges in Reaching the Island of Stability

- Production difficulties: Superheavy elements require high-energy particle accelerators and rare target materials.
- Short lifetimes: Although some isotopes show increased stability, most decay quickly.
- Confirmation challenges: Detecting and characterizing these elements require advanced spectrometry and collision experiments.

Summary

Subatomic particles of the nucleus (protons and neutrons) are called **nucleons**. A **nuclide** is an atom with a particular number of protons and neutrons. An unstable nucleus that decays spontaneously is **radioactive**, and its emissions are collectively called *radioactivity*. Isotopes that emit radiation are called **radioisotopes**. Each nucleon is attracted to other nucleons by the **strong nuclear force**. Stable nuclei generally have even numbers of both protons and neutrons and a neutron-to-proton ratio of at least 1. Nuclei that contain *magic numbers* of protons and neutrons are often especially stable. **Superheavy elements**, with atomic numbers near 126, may even be stable enough to exist in nature.

Discussion question

1. Identify the role of the strong nuclear force in maintaining nuclear stability and explain how it overcomes electrostatic repulsion between protons.
2. Predict what would happen to an element with a neutron-to-proton ratio much higher than the band of stability.
3. Find why elements beyond atomic number 83 (Bismuth) are inherently unstable and eventually undergo radioactive decay.
4. Identify how the concept of magic numbers influences nuclear stability and compare it to the electron shell model in atoms.
5. Predict how the discovery of long-lived superheavy elements could impact our understanding of nuclear physics and stability.
6. Find why the isotope Lead-208 is considered one of the most stable heavy elements, and explain its significance in nuclear science.